Demographic Microsimulator (DEMOS)
Technical Memorandum

OVERVIEW

DEMOS is an agent-based simulation framework which is utilized to evolve population demographics or key lifecycle events. DEMOS modules are designed to capture the interdependencies of short-term and long-term lifecycle events that are essential for downstream transportation modeling. The primary facet of DEMOS is that this model captures the impact of an agent's current state (i.e., education, marital status etc.,) on their future state. This has important consequences on medium- and long-term transportation modeling such as household vehicle transactions, or work location decisions.

The key features of DEMOS are:

- **10+ lifecycle events**: Aging, education participation, marriage/cohabitation, childbirth, separation, mortality, labor force participation, income growth and evolution, and migration modules.
- **Supported by long running panel data**: the behavior modules of DEMOS were estimated from the Pannel Study of Income Dynamics (PSID) data, which contains over 50 years of longitudinal panel data of American families and measures social, economic, vehicle ownership and health factors over the life-course.
- **Capturing model interdependencies**: lifecycle events can influence each other, which allows users to investigate the impacts of demographic trend in population and how it affects other lifecycle events.
- **Flexible simulation structure and modularity**: the sequence of lifecycle events (shown in Figure 1) can be easily adjusted to capture the region trends and the modules can be replaced as desired by user.
- **Application cases**: current application examples include mandatory location choice models and household vehicle transaction.

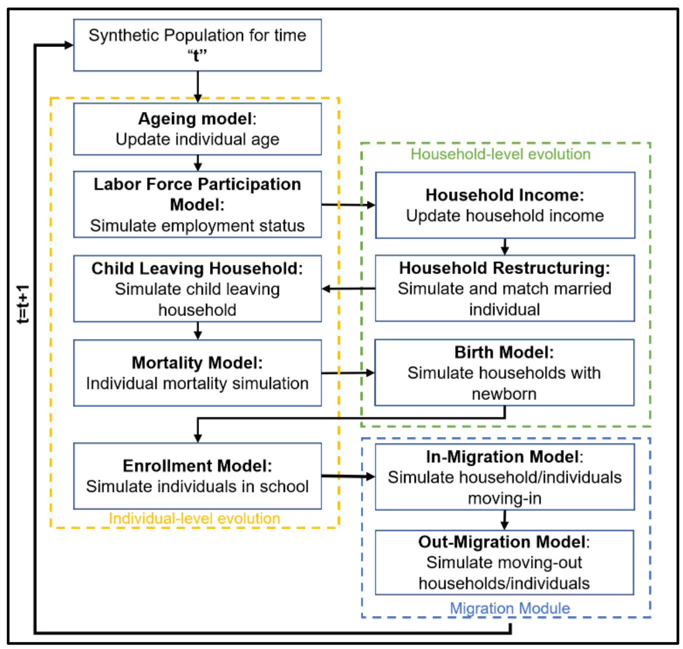

Figure 1. Demographic Microsimulator Sub-Models

This version differs from the LBNL_DEMOS in following ways:

- LBNL_DEMOS is a collaborative effort across multiple research organizations. Core models in DEMOS were estimated by researchers at the National Laboratory of Rockies and Lawrence Berkely National Laboratory, and implemented in UrbanSim framework by researchers from UrbanSim, inc. Using this version requires the UrbanSim environment.

- DEMOS (hosted in this repo) build upon LBNL_DEMOS, with improvements to demographic behavior modules and model calibration methods. It also features optimized code structure and enhanced usability through a centralized configuration file and status reporting. This version can be used independently without requiring additional software dependencies.

DEMOS FRAMEWORK

The overall framework of DEMOS (depicted in Figure 1) consists of three major components: i) migration module, ii) individual-level demographic evolution, and iii) household-level demographic evolution. The demographic evolution process is initiated with a baseline-year (t) synthetic population and advances individuals and households through a host of lifecycle events. Household-, and individual-level characteristics are updated and provided as inputs to subsequent year's (t+1) population evolution. This process is repeated to evolve the population of a study region over a span of 10-30 years.

REQUIREMENTS TO ADOPT DEMOS FOR IMPLEMENTATION

Input Datasets

Model Estimation

DEMOS has the default set of demographic behavior models, which are estimated from the U.S. nationwide PSID data and are currently in discrete choice model formats. Users can adopt the DEMOS default models and calibrate with local data (details in section Model Application) or leverage a state specific version of PSID data (which is free to download and use) to customize DEMOS models to your state.

If your agency would like to adopt DEMOS for your region from scratch with own demographic behavior models, survey data is needed. It would be ideal to have a dataset that capture individual and household level life cycle events that comprise DEMOS. A panel survey dataset would be ideal to capture the dynamics of demographic evolution, but in the absence of that a cross-sectional survey can also be leveraged.

Model Application

The DEMOS requires the following datasets to carry out a multi-year simulation:

- **Baseline synthetic population:** A synthetic population dataset of individual and household agents resulting from users' choice of synthetic population generator. All of the variables currently used by DEMOS models are internally consistent (i.e., pertaining to individual and household characteristics of the synthetic population). The synthetic population will be evolved over a given horizon period in time increments (e.g., 1 year, 2 years, or 5 years) chosen by the analyst.
- **Calibration data:** Observed and future forecasted data for the different DEMOS submodules are required to calibrate the modules. Depending on the demographic events of interests, example calibration data can be population size, number of total households, number of yearly births, number of yearly mortalities, total number of single, married, cohabitating, and divorced individuals, number of students, labor force size, yearly income growth rates. The calibration data can be gathered either from historically observed data such as census data or data maintained by local agencies, or the projected future demographic trends that users are interested in testing.

Output Data

DEMOS generates demographic data for simulated individuals and households, detailing every demographic event shown in Figure 1. Key outputs from DEMOS are person and household tables generated for each simulation step (such as annually), provided as CSV files. The person table includes demographic details for each simulated individual, with each row representing a single agent. The household table, similarly, lists data for all simulated households. These tables use household IDs and person IDs to delineate familial or household relationships. From these outputs, users can derive specific metrics of interest, like the count of households with more than two workers. By default, DEMOS offers aggregated data outputs, including total population size, total number of households, the count of households with children under 18, and statistics on enrollment, as well as the number of employed women and married women within the population.

HOW CAN DEMOS HELP WITH ENHANCEMENT OF TRANSPORTATION PLANNING

- Replacing cross-sectional synthetic populations with a systematically evolved one
  DEMOS can generate a synthetic population that evolves over time based on observed data and behavioral rules. This output can replace the current methods for generating synthetic populations, seamlessly integrating with existing planning workflows. For instance, DEMOS-generated data can serve as inputs for models that simulate choices regarding home and workplace locations, as well as activity-based travel demand modeling.

- Provide Population Composition for Various Demographic Trends
  DEMOS offers the capability to simulate demographic changes under various user-defined scenarios, providing detailed insights into specific population segments of interest. For example, it can simulate population projections within certain optimistic or pessimistic boundaries which can then be used as inputs of scenarios impacted by demographic trends. Furthermore, DEMOS can detail the composition of the population under different scenarios, such as population aging trends, by allowing users to specify parameters for mortality and migration models. One relevant example here is the decline in birth rates during the initial stages of the COVID-19 pandemic followed by a surge in births. The impact of trends such as this on vehicle ownership, home purchase decisions, and work location or telework decisions will all require a population that evolved year or over year compared to two independent snapshots of synthetic population.

- Broad Application Potential
  DEMOS enables a comprehensive representation of mobility behaviors by incorporating life events and trajectories, which are crucial for understanding individual mobility decisions and behaviors. A recent study by Jin et al. (2022) highlighted the application of DEMOS in modeling household vehicle transactions, considering a wide range of demographic and vehicle-specific factors. These include changes in household income, family size, marital status, residential location, and educational attainment, all of which are pivotal in predicting decisions regarding vehicle ownership and turnover.

  Additionally, DEMOS's ability to simulate changes in demographic characteristics can benefit other applications sensitive to such dynamics. This includes the adoption of new technologies (e.g., automated vehicles, on-demand mobility services), teleworking trends, real estate market modeling, and studies on health and disease transmission.